

Stock Portfolio Optimization using Reinforcement learning

Lakshmi Narayanan S.K.

Department of Mathematics, Shiv Nadar Institute of Eminence

ls916@snu.edu.in



Abstract

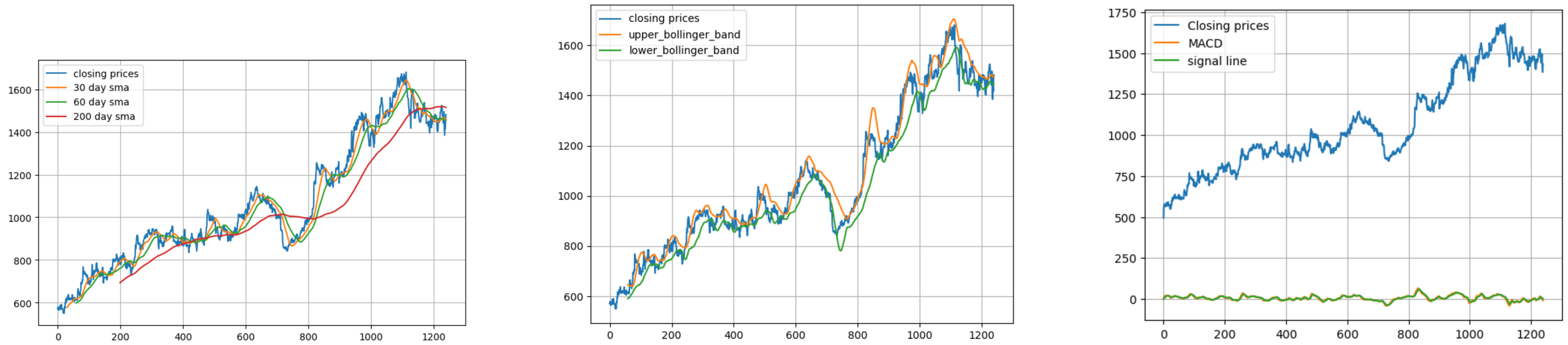
This study presents a reinforcement learning-based model for portfolio optimization, targeting stocks from two sectors of the National Stock Exchange (NSE). The model is driven by a profit-and-loss-oriented reward function and learns trading policies based on stock behavior patterns. To benchmark the performance of the reinforcement learning approach, a traditional technical analysis-based strategy—incorporating indicators such as moving averages, Bollinger Bands, and Moving Average Convergence Divergence (MACD)—is used for comparison. The research further evaluates the effectiveness of various deep reinforcement learning algorithms in trading decisions, offering insights into their comparative performance against indicator-driven strategies.

Data Collection

The stock data for this research is collected from Yahoo finance. Instead of downloading the data from the website as an excel file we use the 'yfinance' library in python to directly access the data. The data primarily contains historical data of Open, High, Low, Close prices for companies along with the dates in which prices were recorded. We have considered to use the price data for the past 5 years of 6 companies from the pharmaceutical sector.

Technical Analysis Indicators

Technical analysis is a method of evaluating statistical trends in trading activity, typically involving price movement and volume. It is used to identify trading and investment opportunities. We have used Moving averages, Bollinger bands and Moving average Convergence Divergence (M.A.C.D)



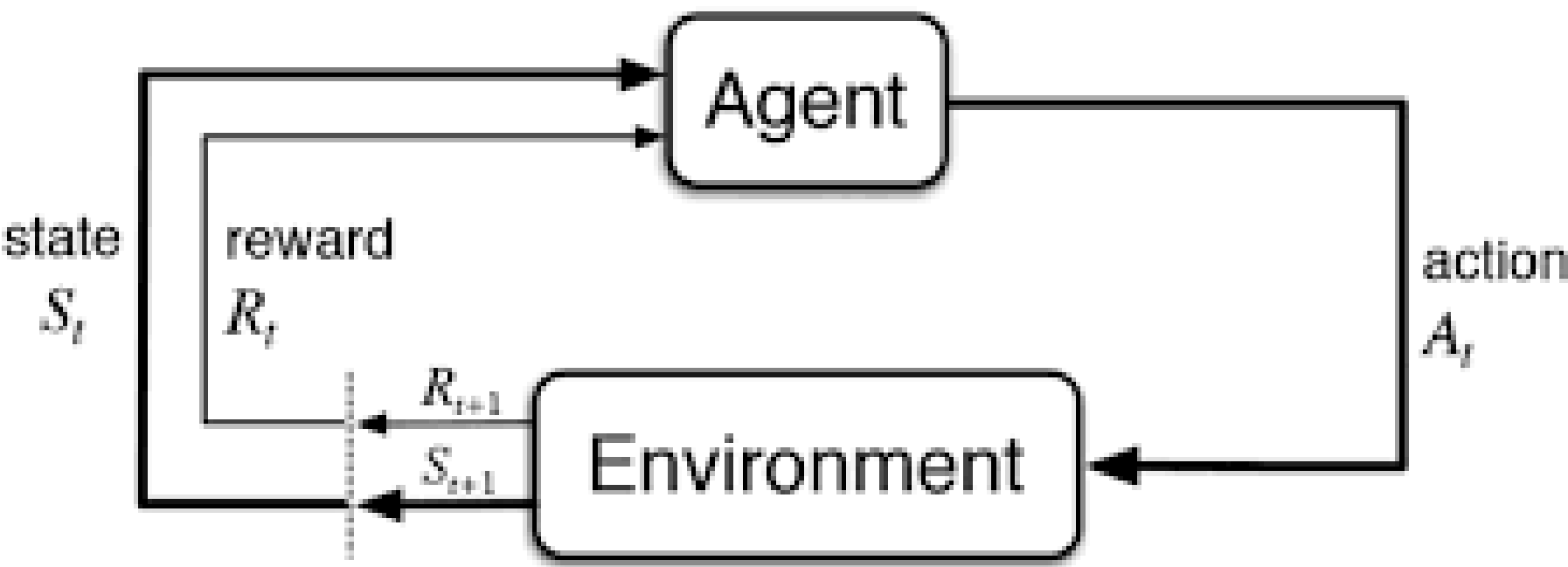
Reinforcement learning Introduction

Reinforcement learning is a subfield of machine learning where an agent learns by interacting with its environment. The agent begins in an initial state and performs an action to transition to a new state. In each state, the environment provides a reward based on the action taken. The objective of a reinforcement learning algorithm is to maximize the cumulative reward as the agent progresses toward the terminal (final) state.

$$\pi(a \mid s) = \text{Prob}(A_t = a \text{ if } S_t = s) \tag{1}$$

$$v_{\pi}(s) = E[\sum_{k=1}^N \gamma^k * R_{t+k+1} | S_t = s] \tag{2}$$

$$q_{\pi}(s, a) = E[G_t | S_t = s, A_t = a] \tag{3}$$



REINFORCE Trading Methodology

The state (S_t) needs to be converted to a tensor that looks like [cash, 0, 0, 0, 0, n1, n2, n3, n4, n5, n6, c1, c2, c3, c4, c5, c6]. Where n1, n2, n3... are the number of stock holdings of the respective companies at that time t and each action is taken to be a vector (integer matrix) as $A_t = [a1, a2, a3, a4, a5, a6]$ where each integer in this vector represents a trading action for that particular stock.

To define the probabilities of actions given a state, we use the softmax function, which is commonly used in classification problems involving more than two classes.

$$\pi(a \mid s, \theta) = \frac{e^{h(s, a, \theta)}}{\sum_b e^{h(s, b, \theta)}} \tag{4}$$
$$h(s, a, \theta) = \theta^T x(s, a)$$

REINFORCE update rule is used to learn and update the policy (trading strategy) after every state.

$$J(\theta) = v_{\pi}(s_0) \tag{5}$$

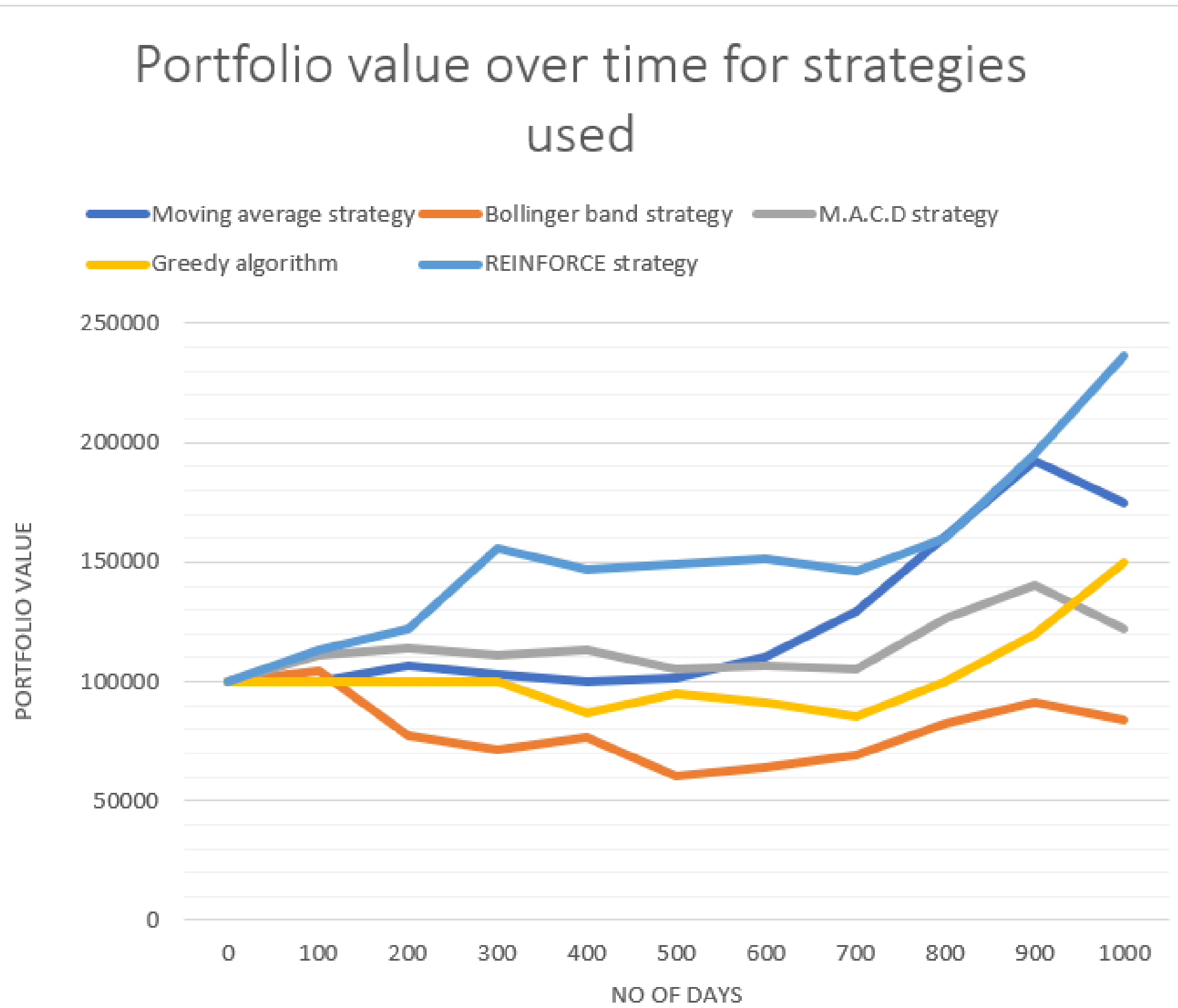
$$\nabla_{\theta} J(\theta) = E_{\pi} \left[q_{\pi}(S_t, A_t) \frac{\nabla_{\theta} \pi(A_t \mid S_t, \theta)}{\pi(A_t \mid S_t, \theta)} \right] \tag{6}$$

$$G_t = \sum_{k=t+1}^T \gamma^{k-t-1} R_k \tag{7}$$

$$\theta \leftarrow \theta + \alpha_t G_t \nabla_{\theta} \log \pi(A_t \mid S_t, \theta) \tag{8}$$

Results

After comparing our REINFORCE algorithm with technical analysis strategies, we can conclude that our algorithm is feasible.



References

- Technical Analysis: The Complete Resource for Financial Market Technicians by Charles D. Kirkpatrick, Julie R. Dahlquist
- Reinforcement Learning: An Introduction, Second edition, Richard S. Sutton and Andrew G. Barto

Acknowledgement

I would like to thank Dr. Charu Sharma for their guidance and support throughout this project. I also appreciate the resources and encouragement provided by the Mathematics department of SNIOE.